

Title— *An Integrated Smart Pond Water Quality Monitoring and Fish Farming Recommendation Aquabot System*

• Topic of interest:

In recent years, Smart fish farming has emerged as a promising approach to enhance the productivity, sustainability, and efficiency of aquaculture by integrating Internet of Things (IoT) and machine learning technologies. In integrated smart pond systems, the monitoring of water quality and fish farming recommendations relies heavily on the Internet of Things (IoT) and machine learning (ML). These systems utilize a variety of sensors to measure critical parameters such as temperature, turbidity, dissolved oxygen, and pH, providing real-time insights for maintaining optimal pond conditions. Collected data is sent to cloud-based repositories for storage and analysed using machine learning algorithms to recommend suitable fish species, optimize feeding schedules, and predict risks such as disease outbreaks. While these technologies show significant potential for improving productivity and sustainability in aquaculture, a major challenge lies in data integration. The data collected from diverse sensors and systems often exist in varying formats and resolutions, making it difficult to combine and process them effectively. This fragmented data hinders the development of unified decision-support systems, which are crucial for delivering accurate and actionable insights to farmers. Additionally, integrating large-scale datasets while maintaining data quality and consistency remains a technical hurdle. Inadequate integration may also lead to incomplete or biased analysis, limiting the reliability of fish species recommendations and predictive risk assessments. Addressing this issue requires robust data fusion techniques, standardized protocols for sensor outputs, and scalable systems capable of managing diverse data streams. By overcoming these challenges, smart pond systems can provide farmers with seamless, reliable, and data-driven solutions, ultimately advancing sustainable fish farming practices.

• Problem of Statement:

Small to medium-scale fish farmers face challenges in achieving accurate fish species recommendations and water quality monitoring due to class imbalance in machine learning models, sensor data variability, and system scalability issues. The existing systems often fail to handle imbalanced datasets, leading to inaccurate predictions, while sensor inconsistencies hinder real-time water quality monitoring [7]. Additionally, many current solutions lack scalability for larger ponds, making them unsuitable for wide-scale adoption, particularly in rural or resource-limited environments. [14] Highlights the limitations of fish species

prediction models without preprocessing techniques like feature scaling and data balancing. [8] Discusses real-time water quality monitoring but lacks automated decision support systems. [10] Demonstrates IoT implementation for aquaculture but lacks scalability and intelligent features. [16] Focuses on sensor-based monitoring but identifies gaps in decision support for fish farming. Fish farmers, particularly in small to medium-scale operations, struggle with implementing effective integrated smart pond water quality monitoring systems that provide accurate fish species recommendations [12]. Current systems often suffer from class imbalance in datasets, leading to poor machine learning model performance and unreliable predictions [9]. Moreover, real-time monitoring is hindered by sensor variability and limited system scalability, making it challenging for farmers in rural or resource-constrained areas to adopt these technologies. This study aims to develop a scalable IoT-enabled smart aquabot system that addresses these challenges by improving sensor accuracy, balancing machine learning datasets using advanced techniques like SMOTE, and delivering precise fish species recommendations. By enhancing these systems, fish farmers can make informed, data-driven decisions, increasing productivity, reducing operational costs, and promoting sustainable aquaculture practices. This problem statement clearly identifies who (fish farmers), what (issues with class imbalance, scalability, and sensor variability), when (during real-time monitoring and fish recommendation), where (small to medium-scale rural fish farms), why (to improve productivity and sustainability), and how (through IoT integration, data balancing, and machine learning enhancements), leading directly to the study's primary research question: How can an integrated IoT-based system optimize fish species recommendations and real-time water quality monitoring for sustainable fish farming?

• **Primary research question:**

How can the recommendation of fish species for sustainable aquaculture be enhanced by the combination of machine learning algorithms and Internet of Things-based water quality monitoring?

• **Objective:**

1. To look into how well machine learning systems forecast appropriate fish species using data on water quality in real time. To achieve this goal, various machine learning models including Random Forest, SVM, and ensemble models will be trained and evaluated in order to determine the optimal strategy for recommending fish species based on variables like pH, temperature, and turbidity.

2. To look into how IoT-based real-time water quality monitoring devices might enhance sustainability and decision-making in fish farming. The assessment of sensor integration for continuous data collecting, the difficulties of real-time data transmission, and the ways in which this data might support more sustainable and effective fish farming methods—especially in off-grid sites that are powered by solar energy—will be the main objectives of this aim.

• **Hypothesis:**

The integration of IoT-based real-time water quality monitoring and machine learning for fish species recommendation will significantly improve the productivity and sustainability of fish farming by providing accurate, data-driven decisions for water quality management and species selection. This hypothesis reflects the expectation that the proposed system will enhance fish farming efficiency through real-time monitoring and automated decision support for selecting suitable fish species based on water quality.

• **Literature Review:**

The integration of emerging technologies such as the Internet of Things (IoT), machine learning (ML), and robotics has revolutionized various sectors, including agriculture and aquaculture. In the context of smart fish farming, the application of IoT for real-time water quality monitoring and ML for fish species recommendations has gained significant attention in recent years. Several studies have investigated the benefits and challenges of implementing these technologies in fish farming, with a focus on improving productivity, minimizing costs, and promoting sustainability. This literature review explores various research articles related to IoT-based water quality monitoring systems, fish species recommendations, and the integration of machine learning and renewable energy in smart aquaculture.

IoT-based Water Quality Monitoring in Fish Farming:

Water quality is a crucial factor in fish farming, as it directly influences fish health, growth, and survival rates. Monitoring water quality parameters such as temperature, pH, turbidity, dissolved oxygen (DO), and salinity is essential for maintaining optimal conditions for fish. However, traditional methods of water monitoring are often labour-intensive and time-consuming, making them less effective in ensuring real-time responses to sudden changes in water quality. [4] developed an IoT-based water quality monitoring system that collects data from multiple sensors and transmits it to a cloud database for analysis. The system uses a Raspberry Pi microcontroller with an in-built Wi-Fi module to serve as the gateway for

communication. The data is processed and sent to the farmer's device via a mobile application if water quality parameters fall outside the optimal ranges. While the system successfully monitors water quality, it does not provide any recommendation for selecting fish species based on water quality indicators, a key feature in enhancing the productivity of fish farming. In another study, [5] proposed a cloud-based water monitoring system comprising five phases: data extraction, cloud platform, database storage, reports, and predictions. Their system uses sensors to gather data, which is then transferred to a cloud platform for storage and processing. The recorded data are shown on dashboard reports, and predictions are made based on historical data to forecast future water quality parameters. Although the system provides valuable water quality data and predictions, it lacks an intelligent recommendation feature for fish species, which could have a significant impact on farming efficiency.

Machine Learning for Fish Species Recommendations:

The recommendation of appropriate fish species based on water quality is an important aspect of smart fish farming. The use of machine learning algorithms for analysing water quality data and recommending suitable fish species can help farmers make informed decisions and optimize their farming practices. [3,14] developed a model to predict suitable fish species based on water quality parameters such as pH, temperature, and turbidity. They applied machine learning algorithms, including Random Forest (RF), k-Nearest Neighbours (KNN), and J48, to analyse real-time pond data and achieved 88.48% accuracy with the RF model. This study demonstrates the potential of machine learning in making fish species recommendations based on water quality, but it primarily focuses on the accuracy of the models rather than the integration of these models into a fully automated system for real-time decision-making.

IoT and Renewable Energy Integration in Smart Aquaculture:

Sustainability is a key concern in smart aquaculture, and the integration of renewable energy sources, such as solar power, can play a significant role in reducing the carbon footprint of fish farming operations. Solar-powered systems are particularly attractive in remote areas, where access to electricity may be limited. [7] Presented a solar-powered aquaponics system that incorporates a water-recirculating system for plant and fish growth, alongside a control system for monitoring and regulating water temperature. While the system provides a multidisciplinary approach to aquaponics, it lacks an intelligent decision support system for choosing fish species based on water quality parameters. The integration of solar power with water monitoring could improve the sustainability and cost-effectiveness of the system, but the absence of data-driven

decision support highlights a research gap in integrating renewable energy with machine learning-based decision systems. Similarly, [12] developed a solar-powered aquaponics system for monitoring water quality parameters like pH and temperature. The system uses sensors and GSM to notify farmers if parameters deviate from the ideal range. While this approach benefits from solar power, it lacks automated decision-making capabilities for recommending specific fish species, which limits its effectiveness in optimizing farming practices.

Smart Pond Management Using Robotics:

The use of robotics in smart aquaculture is another emerging trend that aims to automate the monitoring of water quality and the collection of data from remote locations in the pond. Robotic systems can help reduce the labour required for monitoring and enhance the overall efficiency of fish farming operations. [2,6] proposed an IoT-based fish farming and tracking control system that uses an intelligent management module for continuous water quality monitoring. The system integrates various sensors to collect water quality parameters, which are then analysed for forecasting future conditions. While this approach focuses on intelligent management of water quality, it does not incorporate a recommendation system for selecting fish species. Additionally, the study does not mention the use of renewable energy or robotics to automate the monitoring process.

Research Gaps and Future Directions:

While numerous studies have explored the application of IoT and machine learning in fish farming, several gaps remain in the literature. First, the integration of intelligent decision support systems, particularly for recommending fish species based on real-time water quality data, is limited. Most existing systems [9-13] lack automated decision-making capabilities, which hinders the ability of farmers to make timely, data-driven decisions. Second, while many studies incorporate IoT-based water quality monitoring systems, there is a need for systems that combine IoT, machine learning, and robotics to provide a comprehensive solution for smart aquaculture. The integration of renewable energy sources, such as solar power, also remains underexplored, especially in the context of sustainable and self-sustaining systems for fish farming. Lastly, real-time predictive analytics for water quality is an area that requires further research. Existing systems often focus on historical data analysis and predictions based on past conditions, but there is a need for systems that can predict and address potential water quality issues as they occur. This would enable farmers to take proactive measures, reducing the risk of fish mortality and improving overall farm productivity.

• Design:

A combination of qualitative and quantitative research method can be used for this project.

1. Quantitative Component: This aspect focuses on collecting numerical data through sensors integrated into the smart pond system. These sensors will monitor various environmental factors such as water quality, temperature, and pH levels, providing valuable data for system performance analysis.
2. Qualitative Component: This component aims to capture the user experience and perception of the Aquabot system. Through surveys, interviews, or feedback sessions, qualitative methods will help understand how users interact with the system, their satisfaction levels, and any improvements they suggest.

• Research Variables:

Independent Variables:

These are the variables that influence or affect the outcome. In the study, the independent variables are:

1. Water Temperature: The temperature of the pond water that affects fish survival and growth.
2. pH Level: The acidity or alkalinity of the water, which influences fish health.
3. Turbidity: The clarity of the water, which affects fish feeding and overall health.
4. Solar Power Utilization: This variable is concerned with whether the system utilizes solar power as a sustainable energy source to drive the IoT devices, sensors, and robotic components for water quality monitoring.

Dependent Variable:

This is the variable that depends on or is influenced by the independent variables. In our study, the dependent variable is:

1. Fish Species Recommendation – The type of fish suggested based on the water quality parameters (temperature, pH, and turbidity).
2. Fish Growth Rate: The rate at which fish grow in the monitored pond is an indirect dependent variable influenced by water quality and the appropriateness of the selected fish species.

• **Scientific Methods:**

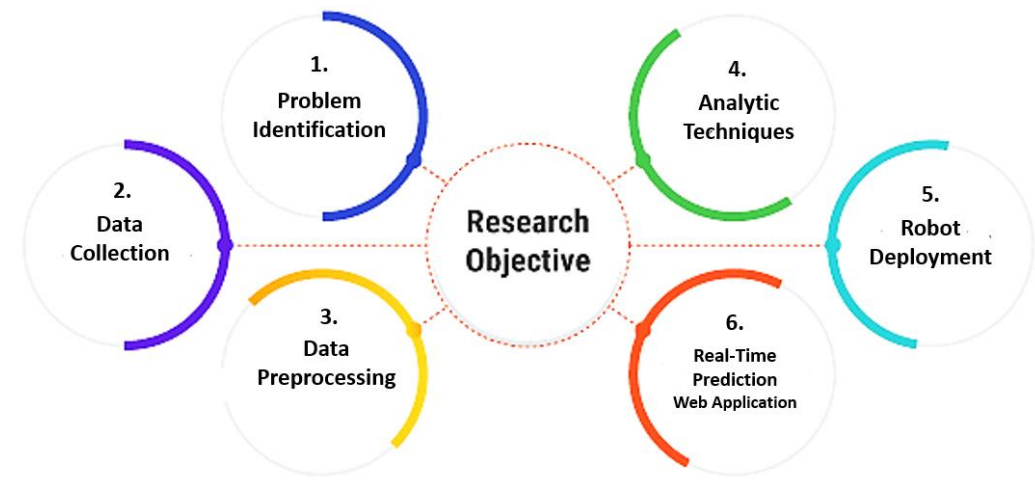


Figure: 1: Step by step process to achieve the objectives

The proposed method is basically an Application Based Research Method. Figure 1 shows the diagram of the processes that is needed to achieve the objective of this project. Including:

1. Problem Identification:

To design a solar power-integrated smart aqua robot that operates autonomously in aquatic environments. Develop a real-time prediction system for the robot's performance and environmental factors using machine learning and big data analysis.

2. Data Collection:

Temperature, water quality (pH, salinity, turbidity), motion, battery status, solar panel performance.

3. Data Preprocessing:

After collecting the data we need to preprocess the data and make it appropriate for our analysis to fulfil the objectives.

4. Analytic Techniques:

Here we can use Machine learning to analysis the collected data and to apply it for further predictions.

5. Robot Deployment:

After analysis and training the machine learning, now we can deploy our robot that can collect the real time data from the pond and save it to cloud for prediction.

6. Real time Prediction Web Application:

One of the most important parts of this project is to show real time prediction of fish species in a web application.

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